

Monday

10-11 in person + Zoom (link will be on Classroom)

Inversions (Discrete symmetries)

Transformation Π inversion

Finite group with 2 elements $\Pi, \mathbb{I}$

parity: $\vec{x} \rightarrow -\vec{x}$ $\Pi \psi(\vec{x}, t) = a \psi(-\vec{x}, t)$

$$\underbrace{\Pi \Pi}_{4\mathbb{I}} \psi(\vec{x}, t) = a \Pi \psi(\vec{x}, t) = a^2 \psi(\vec{x}, t)$$
$$\Rightarrow \Pi = \Pi^{-1}$$

$$\langle x | \Pi | x' \rangle = \langle x | -x' \rangle = \delta(x + x')$$

$$\langle x | \Pi^\dagger | x' \rangle = \langle -x | x' \rangle = \delta(-x - x') = \delta(x + x')$$

$$\Rightarrow \Pi^\dagger = \Pi \Rightarrow \Pi^\dagger = \Pi = \Pi^{-1} \Rightarrow \Pi \text{ Hermitian}$$

$\Rightarrow$ no need for generator of Π

Inversion Π $\Pi \psi_1 = \mathbb{I}_1 \psi_1$

$$\Pi(\psi_1 \psi_2) = \Pi \psi_1 \Pi \psi_2 = \mathbb{I}_1 \psi_1 \mathbb{I}_2 \psi_2 = (\mathbb{I}_1 \mathbb{I}_2) \psi_1 \psi_2$$

$$\mathbb{I}_{(1+2)} = \mathbb{I}_1 \cdot \mathbb{I}_2 \quad \text{multiplicative}$$

In classical mechanics: parity related to rest frame $\psi(x)$ of particle

classical mechanics: $E = \left(\frac{p^2}{2m} \right) \rightarrow$ classical limit of

Relativ.

$$E = m + K$$

$$E^2 = p^2 + m^2$$

In QFT parity defined for all particles, also massless.

Elementary particles:

intrinsic parity a pure convention

$$\text{fermions} \left\{ \begin{array}{l} \text{leptons } \{ e^-, \mu^-, \tau^-, \nu \} \\ \text{quarks} \end{array} \right. \quad P = +1$$

$$\text{anti-fermions} \left\{ \begin{array}{l} \text{anti-leptons: } e^+, \mu^+, \tau^+, \bar{\nu} \\ \bar{q} \end{array} \right. \quad P = -1$$

$$\text{elementary bosons and: anti-bosons} \quad P = +1$$

Composite states

f: fermion

b: boson.

$$f_1 \bar{f}_2 \quad P = P_1 P_2 (-1)^L = (+1)(-1)(-1)^L = (-1)^{L+1}$$

L : angular momentum

proton: uud

$$P = (+1)(+1)(+1)(-1)^L$$

proton has $L=0 \Rightarrow P_{\text{proton}} = +1$

neutron: udd

$$L=0 \Rightarrow P_{\text{neutron}} = +1$$

Pion: $\pi^+ = (u\bar{d})$

$$P = (+1)(-1)(-1)^L$$

$L=0 \Rightarrow P_{\pi^+} = -1$

$$\pi^- = \bar{u}d \Rightarrow P_{\pi^-} = -1$$

photon γ :

$$P_{\gamma} = -1$$

Different quantities under parity $\mathbb{P}$:

scalar S : $\mathbb{P} S = S$ invariant

vector V : $\mathbb{P} V = -V$

$$\mathbb{P} \vec{x} = -\vec{x} \quad \mathbb{P} \vec{p} = -\vec{p}$$

$$S = \vec{p} \cdot \vec{x}$$

$$\mathbb{P} S = \mathbb{P}(\vec{p}) \cdot \mathbb{P}(\vec{x}) = (-\vec{p}) \cdot (-\vec{x}) = \vec{p} \cdot \vec{x}$$

pseudo vector: $\mathbb{P} A = A$

$$\vec{L} = \vec{x} \wedge \vec{p}$$

$$\mathbb{P} \vec{L} = \mathbb{P} \vec{x} \wedge \mathbb{P} \vec{p} = (-\vec{x}) \wedge (-\vec{p}) = \vec{x} \wedge \vec{p} \equiv \vec{L}$$

pseudo scalar:

$$\mathbb{P} p_s = -p_s$$

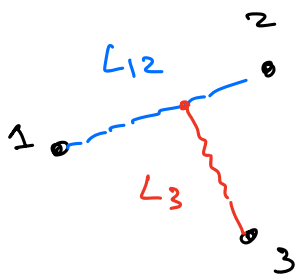
$$\text{helicity} = \frac{\vec{p} \cdot \vec{s}}{|\vec{p}| |\vec{s}|}$$

$$\mathbb{P} \text{helicity} = -\text{helicity}$$

System of particles

$$\psi = \psi_1 \psi_2 \psi_3$$

$$\mathbb{P} \psi = \underbrace{P_1 P_2 P_3}_{\text{intrinsic parities}} \mathbb{P} \psi_{\text{space}}$$



$$P = P_1 P_2 P_3 (-1)^{L_2 + L_3}$$

L_3 : ang. mom of 3 with respect to center of mass of 1-2 syst.

parity conservation

EM ☒
weak ☒
Strong ☒

Experiment by Wu 1957
 $^{60}\text{Co} \rightarrow ^{60}\text{Ni}^* + e^- + \bar{\nu}_e$

$\mathbb{C}$ parity (Charge Conjugation)

$\mathbb{C}$: invert the electric charge

$$\mathbb{C} e^- \rightarrow e^+$$

$\vec{x}, \vec{p}, \vec{s}$ are unchanged
under $\mathbb{C}$
only electric charge.

$$\mathbb{C} \psi = a \psi$$

$\Rightarrow$ only neutral particles candidates to be eigenstates

$$\mathbb{C}^2 \psi = \mathbb{C} [\mathbb{C} \psi] = \mathbb{C} [a \psi] = a^2 \psi$$

$$\Rightarrow a^2 = 1 \Rightarrow a = \pm 1$$

n, γ, π^0, ν ? are eigenstates.

γ : eigenstate of $\mathbb{C}$ $C_\gamma = -1$ from EM

$$\underline{n \stackrel{?}{=} \bar{n}}$$

$$\left. \begin{array}{l} n \rightarrow e^- + \bar{\nu}_e + p \\ \mathbb{C} \bar{n} \rightarrow e^+ + \nu_e + \bar{p} \end{array} \right\} \text{Can be distinguished} \\ \text{with magnetic field}$$

$$\Rightarrow n \neq \bar{n}$$

$$\underline{\nu \stackrel{?}{=} \bar{\nu}}$$

still in discussion
open question

Dirac Neutrino $\nu \neq \bar{\nu}$
Majorana Neutrino $\nu \equiv \bar{\nu}$

$$n = udd \xrightarrow{\mathbb{C}} \bar{u} \bar{d} \bar{d} \quad \text{different quark content}$$

In QFT: $C_{f\bar{f}} = -1$ opposite charge conjugation
for f and $\bar{f}$

$C_{b\bar{b}} = +1$ same $\mathbb{C}$ for elem. boson
and anti-bosons.

$\mathbb{C}$ useful for system of particles (decays)

$$\pi^+ \pi^- \xrightarrow{\mathbb{C}} \pi^- \pi^+$$

$$\pi^+ \pi^- \pi^0$$

$$\pi^+ \pi^- \pi^+ \pi^-$$

$$\pi^0 \pi^0 \pi^0$$

$$K^+ K^-$$

} potential
eigenstates.